

HW3 Section 2 — Real-World Problem: Weather Analytics via Hadoop MapReduce and gRPC

Problem continued from Homework 2, Q8: Large-Scale Weather and Environmental Data Analytics. Section 2 re-implements the same analytics problem under two different distributed-system paradigms — Q1 under a batch MapReduce model, Q2 under a real-time streaming model — to study how the choice of paradigm affects design, communication, scalability, and performance.

This report covers both questions. Source code, full setup/run instructions, and additional detail live in each question's own README (`Q1_MapReduce/README.md` , `Q2_gRPC/README.md`); this document consolidates the architecture, methodology, and results from both into one place.

Reading guide: results are labeled either **[Measured]** (a real number from an actual run, never invented) or **[Interpretation]** (our explanation of why a measured result looks the way it does). Sections also flag **[Required]** vs. **[Optional]** against the assignment's actual wording, and **[Real Hadoop]** vs. **[SLURM-based substitute]** wherever execution mode matters.

1. Problem and Background

HW2 Q8 processes weather/environmental sensor records — `timestamp station_id temperature humidity pressure rainfall wind_speed` — and computes 19 output fields: global statistics (count, averages, min/max for temperature/humidity/pressure), total/max rainfall, average/max wind speed, an extreme-temperature event count, the hottest and coldest single measurement, the busiest 60-second interval, and the top-K stations by measurement count. Tie-breaking rules (smaller timestamp, then smaller station ID) and the input/output format are fixed by HW2 and preserved exactly in both Q1 and Q2 here — neither implementation changes the analytics themselves, only how the computation is distributed.

Both implementations share the same dataset generator (`generate.cpp` , HW2's own, unmodified) and the same correctness oracle (`weather_seq.cpp` , HW2's unmodified sequential reference) — this is what makes "final output matches a correct sequential implementation" checkable at all, and what keeps Q1 and Q2 directly comparable to each other and to HW2's own MPI implementation.

2. Q1 — Batch Data Analytics using Hadoop MapReduce

2.1 Required environment and what actually ran [Real Hadoop vs. SLURM-based substitute]

The assignment specifies Apache Hadoop 3.3.6, YARN, and C++ via Hadoop Streaming. This project's `mapper.cpp` / `reducer.cpp` are written exactly to that contract — standalone C++ executables communicating with the framework purely through `stdin/stdout` — and `run_hadoop.sh` is a complete, ready-to-run real Hadoop Streaming submission script.

Real Hadoop/YARN was not executed for the results in this report. The RCE cluster's Hadoop/YARN environment has a known issue: HDFS is reachable, but YARN's ResourceManager resolves to a placeholder address (`0.0.0.0:8032`) and cannot schedule jobs — confirmed directly on the account (`yarn node -list` fails with Hadoop's own "endpoint configuration is wrong" diagnostic). The course's own instructions acknowledge this environment issue and allow a SLURM-based script as the interim substitute. All Q1 results in this report are therefore from that **SLURM-based substitute pipeline** (`run_local_pipe_test.sh` / `benchmark.sh`) — labeled as such throughout, never presented as Hadoop timings. `run_hadoop.sh` remains ready to use once real Hadoop/YARN access is restored.

2.2 Mapper design

The mapper reads only data records from `stdin` (never the `N K S` header — stripping that is the driving script's job). It accumulates state across its **entire** assigned split before emitting anything, mirroring the accumulation pattern already used in HW2's own MPI ranks, and emits three kinds of `key<TAB>value` lines at end-of-input:

Key	Value	Purpose
<code>GLOBAL</code>	21 fields: count, 5 Kahan sums, min/max bounds, extreme count, hottest/coldest candidate	This mapper's whole-split contribution to every global statistic
<code>INTERVAL_<id></code>	local count	This mapper's contribution to that 60-second interval
<code>STATION_<id></code>	count, sum_temp, sum_rainfall	This mapper's contribution to that station

An empty split emits nothing. Kahan (compensated) summation is used throughout to avoid floating-point drift across many additions, matching HW2's own approach exactly.

2.3 Reducer design and why only one extra step is needed

`GLOBAL` is a single constant key, so — regardless of how many reduce tasks run — every mapper's `GLOBAL` line always routes to the same one reducer instance (Hadoop's partitioner is a deterministic

function of the key alone). That reducer therefore sees every partial and finishes all 15 global statistics completely, printing them directly in final format. `STATION_<id>` and `INTERVAL_<id>` each route consistently to one reducer per distinct ID, so each reducer produces a **complete** total for the IDs it owns — but with more than one reduce task, different IDs can land on different reducers, so no single reducer sees every station or every interval.

Finding the single busiest interval and ranking the top-K stations therefore needs a view across all reducers' output. Rather than adding a second real MapReduce stage, this is resolved by `finalize.py`, a small non-distributed script — justified because the data involved is provably small regardless of N: interval IDs are capped at 1,441 by the dataset's fixed 24-hour window, and station IDs are capped at S (a small, explicit dataset parameter). A second Hadoop job over data this size would add framework overhead without addressing anything genuinely large.

2.4 Correctness — [Measured]

Verified against `weather_seq_reference` (HW2's unmodified sequential program) on the identical dataset, via strict byte-for-byte `diff`, on:

- Local machine (macOS, ARM64): all test datasets, including hand-built adversarial tie-break cases (2-way/3-way ties on hottest/coldest/busiest-interval/top-K) and generated datasets up to N=1,000,000 — all passed.
- Real RCE hardware (job 98604, x86_64, single node, via the SLURM-based substitute path): passed.

One confirmed, explained discrepancy at N=100,000 — `AVERAGE_WIND_SPEED` differs from the oracle by exactly one unit in the 6th decimal (oracle: `60.115214` ; pipeline: `60.115213`) on real RCE hardware, at every mapper-chunk count tested. Root cause, verified with exact rational arithmetic (`Fraction / Decimal` , no floating point) on the actual dataset: the true average is exactly `60.1152135` — a genuine floating-point halfway point. `kahan.hpp` 's `KahanSum::value()` narrows its internal `long double` accumulator to `double` before the mapper writes it out as text (the Hadoop Streaming stdin/stdout contract is a `double` -precision text protocol), so the reducer's final average division happens in `double` arithmetic, while the oracle stays in `long double` until its very last step. Different rounding paths landing on opposite sides of one exact halfway point is expected at *some* N — this is not a logic bug, and the same underlying dataset value independently reproduces the identical tie in Q2 (Section 3.8 below), with a different root cause but the same phenomenon.

2.5 Benchmark methodology [SLURM-based substitute, single node]

`benchmark.sh` builds the four C++ binaries once, then times only the pipeline execution itself (split → mapper × M → shuffle/sort → reducer → `finalize.py`) — deliberately *not* using `run_local_pipe_test.sh` 's own rebuild-every-invocation behavior, which would add a fixed ~1.5s compile cost to every configuration and swamp the real timing differences at small N. Every configuration also runs a full correctness check against the oracle; a measured time is recorded

regardless of the correctness verdict (so N=100,000's confirmed-benign tie doesn't discard real timing data), and a genuinely failed run is recorded as `FAILED`, never a fabricated number.

Important limitation, stated plainly: under this substitute, mapper chunks run **sequentially on one core** — there is no reducer-count knob either (the reducer is always a single process over the whole shuffled stream). "Mapper chunk count" is therefore the only available parallelism-like knob, and it does not represent genuine multi-core/multi-node execution the way a real Hadoop cluster's mapper tasks would. This is a property of the permitted substitute, not a design choice made by this project.

2.6 Benchmark results — [Measured], job 98781, node06, x86_64, single node

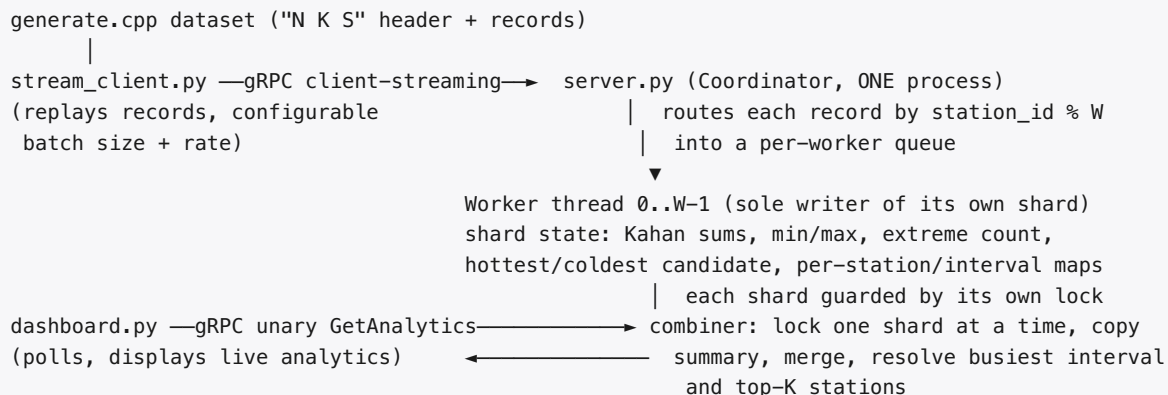
N (records)	M=1	M=2	M=4	M=8
1,000	0.0478 s	0.0318 s	0.0342 s	0.0387 s
10,000	0.0412 s	0.0466 s	0.0554 s	0.0672 s
100,000	0.1313 s*	0.1347 s*	0.1476 s*	0.1731 s*
1,000,000	0.9965 s	1.0033 s	1.0139 s	1.0431 s

* Correctness verdict FAIL at every M — the confirmed benign precision tie from Section 2.4, not a timing anomaly; the times themselves are real and consistent with the surrounding trend.

[Interpretation] Time scales with N as expected. More mapper chunks make things slightly *slower*, not faster (consistent trend, not noise, at N ≥ 10,000) — because each additional chunk is a separate process launch plus more `sort` input, with no real parallel execution across chunks on this substitute (unlike a real Hadoop cluster, where mapper tasks run concurrently across nodes/cores).

3. Q2 — Real-Time Data Analytics using gRPC

3.1 Architecture



One coordinator process with an in-process worker thread pool — not separate gRPC services per worker. This satisfies the assignment's "multiple workers" requirement while keeping the only two network boundaries (client→coordinator, dashboard→coordinator) on a single, well-understood gRPC surface.

3.2 .proto interface

At minimum the assignment requires streaming ingestion and queries for current analytics. The interface provides:

- `StreamRecords(stream IngestMessage)` returns `(IngestSummary)` — client-streaming; the first message is a header (N/K/S), the rest are record batches; returns only after every record sent has been applied (a drain barrier), so a query issued right after this returns is guaranteed to see the whole stream.
- `GetAnalytics(AnalyticsRequest)` returns `(AnalyticsSnapshot)` — unary; safe to call at any time, including mid-ingestion. Batch size is carried inside the message itself, making "streaming granularity" a single tunable rather than two separate knobs.
- `Reset(ResetRequest)` returns `(ResetResponse)` — reconfigures worker count/partition strategy without restarting the process; used by the correctness/benchmark scripts, not part of the required minimum interface.

3.3 Worker state and record distribution

Each worker owns a shard of state (`ShardState`): Kahan sums, running min/max, extreme-event count, a running hottest/coldest candidate, and per-station/per-interval maps for its shard. Two record-distribution strategies were implemented and compared, satisfying the assignment's explicit requirement to investigate this dimension:

- **Station-based partitioning** (default): `station_id % W`. Every record for a given station always lands on the same worker, so per-station sums are built in file order — same order as the sequential reference, byte-identical results guaranteed by construction.
- **Round-robin batch partitioning**: whole batches assigned to workers in rotation. Balances load more evenly, but a station's records can be split across multiple workers, exercising the cross-worker station-merge path at query time.

3.4 Concurrency and locking

One bounded queue per worker; only that worker's own thread ever writes to its shard, so there are no write-write races. A query locks **one worker at a time**, copies a snapshot, releases, and moves to the next — never two locks held simultaneously, so deadlock is structurally impossible, and ingestion is never blocked for longer than one shard's copy takes. Every snapshot is internally consistent (total count equals the sum of per-station counts equals the sum of per-interval counts) but is not guaranteed to be an exact instantaneous prefix of the stream if workers are at slightly different points — an

accepted, documented property, since the assignment asks for "a useful view of the current state," not point-in-time atomicity across shards.

3.5 Dashboard — how it satisfies Q2's CLI dashboard requirement

`dashboard.py` is a CLI-only client (no web interface) with two modes:

- **Live mode** (default): polls `GetAnalytics` every `--interval` seconds (default 1.0 s) and redraws an ingestion-progress header — records applied/expected with a percentage and progress bar, backlog, active/completed stream counts, worker count and partition mode, and one line per worker (records applied, queue depth, busy time) — followed by the full 19-line analytics report in the same field order and precision as the correctness oracle.
- **--once mode**: a single query, printing only the report lines (no debug output), used by the correctness scripts and for quick manual checks.

All of this comes from a single `AnalyticsSnapshot` returned by one `GetAnalytics` call per refresh — the dashboard holds no state of its own between polls. This was verified against a live 1,000,000-record stream: the progress bar visibly advances and `TOTAL_MEASUREMENTS` increases in real time as data arrives, confirmed both in local testing and in a real two-node run on RCE (server on one physical node, dashboard and streaming client on another — see Section 3.9).

3.6 Experimental methodology [Required vs. Optional]

The assignment requires, at minimum, investigating the effect of worker count and the way records are streamed, and separately lists record distribution strategy and concurrent query clients as factors to investigate. Three experiments cover all of these:

Experiment	Varies	Fixed	Satisfies
E1	Worker count $W \in \{1,2,4,8\} \times$ partition strategy	$N=1,000,000$, batch=1,000, unlimited rate	[Required] worker count; record distribution strategy
E2	Batch size $\in$ $\{1,10,100,1,000,10,000\}$	$N=100,000$, $W=4$, station partitioning	[Required] streaming/ message granularity
E3	Concurrent query clients $\in$ $\{0,1,4,16\}$	$N=1,000,000$, $W=4$, station partitioning, batch=1,000	Listed factor: concurrent queries

Each configuration in every experiment also runs a full correctness check against the oracle before its timing is recorded — a run is never reported as fast without its correctness verdict also being checked. **[Optional, not run]**: streaming rate as its own standalone experiment (the `--rate` mechanism itself is implemented and used elsewhere) and station-count/load-skew experiments — both are listed as "useful measurements may include," not required.

3.7 Latency, throughput, and end-to-end time — methodology, kept distinct

These three are measured differently and should not be conflated:

- **Query latency:** measured client-side, immediately before and after a single `GetAnalytics` call (`t0 = time.perf_counter() → RPC → time.perf_counter() - t0`). Pure round-trip time for one query, from the query client's perspective. Reported as p50/p95/p99 over however many queries completed during a run.
- **Throughput / end-to-end processing time:** measured on the streaming client, from immediately before the `StreamRecords` call is issued to immediately after it returns (which only happens once the server's drain barrier confirms every record was applied). `end_to_end_s = t_ack - t_start`; `throughput = records_received / end_to_end_s`. These are two views of the same measured interval, not two different experiments.
- These are measured on **different RPCs entirely** — query latency on `GetAnalytics` calls made by separate query-client threads running *concurrently with*, not as part of, the ingestion timing.

3.8 Correctness — [Measured]

- Local (macOS, ARM64): the full test matrix — 4 hand-built adversarial datasets plus generated datasets up to N=1,000,000, sweeping worker counts and both partition strategies — passed 248/248.
- Real RCE hardware (job 98751, x86_64, single node): 244/248 passed. The 4 failures are all `AVERAGE_WIND_SPEED` at N=100,000, off by exactly one unit in the 6th decimal, at every worker count tested. Exact rational verification (`Fraction / Decimal`) on the real RCE dataset confirms the true value is `60.1152135` — the identical exact halfway point found independently in Q1 (Section 2.4) on the same underlying dataset. Root cause here: Python's `float` is a C `double` ; the oracle's Kahan sum runs in `long double` on x86_64 (confirmed 80-bit — this cannot occur on the ARM64 dev machine, where `long double == double`). Different summation precision landing on opposite sides of the same exact tie — not a logic bug, and the same dataset produces the identical numeric tie independently in two unrelated implementations (Python double-precision Kahan here, a mapper→reducer text round-trip in Q1), which is itself strong evidence this is a genuine floating-point property of the data, not a coincidence or a shared bug.

3.9 Benchmark results — [Measured]

E1 — worker count (job 98780, single node, 3 reps, means shown):

W	Station-based throughput	Round-robin throughput	Server CPU utilization
1	522,358 records/s	531,249 records/s	~102.0%
2	478,272 records/s	529,478 records/s	~101.9%
4	456,882 records/s	534,018 records/s	~101.9%
8	431,689 records/s	536,155 records/s	~102.0%

[Interpretation] Server CPU utilization stays pinned at roughly one core regardless of worker count — Python's Global Interpreter Lock (GIL) means only about one core's worth of work happens concurrently no matter how many worker threads exist, confirmed directly rather than assumed. Station-based partitioning declines noticeably with more workers (–17%, 1→8) because per-record routing does real Python-level work that scales with thread-switching overhead; round-robin batch partitioning stays flat because it routes whole batches with less per-record overhead.

E2 — streaming granularity (job 98780, N=100,000, W=4, 3 reps, means shown):

Batch size	Throughput	Query latency (p95)
1	22,193 records/s	2.18 ms
10	100,597 records/s	1.91 ms
100	317,983 records/s	2.69 ms
1,000	454,893 records/s	7.05 ms
10,000	467,802 records/s	31.15 ms

[Interpretation] Throughput scales roughly 20× from batch size 1 to 1,000 as per-message protocol overhead is amortized over more records, then nearly plateaus. Query latency stays low and stable through batch size 100, then rises sharply — a single very large batch occupies a worker's lock for noticeably longer, a real, measured granularity/latency trade-off. Every row at N=100,000 reports a correctness FAIL — the same confirmed benign precision tie from Section 3.8; included for completeness, not a new finding (it does not depend on batch size).

E3 — concurrent query clients (job 98780, N=1,000,000, W=4, 3 reps, means shown):

Concurrent clients (Q)	Ingestion throughput	p50	p95	p99
0	454,301 records/s	—	—	—
1	371,100 records/s	0.25 ms	0.30 ms	7.58 ms
4	257,879 records/s	0.70 ms	9.91 ms	13.02 ms
16	249,124 records/s	2.83 ms	42.15 ms	47.78 ms

[Interpretation] Ingestion throughput drops by nearly half (0→16 clients) and query latency percentiles rise sharply in step — both consequences of ingestion and queries sharing the same process's GIL. This is the clearest evidence that the single-process design trades query/ingestion isolation for simplicity.

Single-node vs. cross-node (job 99451, server and client on physically separate RCE nodes, same E1 sweep, all 8 configurations passed correctness):

W	Station-based (single-node → cross-node)	Round-robin (single-node → cross-node)
1	522,358 → 404,275 records/s (−22.6%)	531,249 → 409,256 records/s (−23.0%)
2	478,272 → 367,233 records/s (−23.2%)	529,478 → 403,689 records/s (−23.8%)
4	456,882 → 339,469 records/s (−25.7%)	534,018 → 411,409 records/s (−23.0%)
8	431,689 → 320,916 records/s (−25.6%)	536,155 → 403,241 records/s (−24.8%)

[Interpretation] Real inter-node network latency has a consistent, measurable cost — cross-node throughput is ~23–26% lower than single-node across every worker count and both partition strategies, while the shape of each trend (station declining with W, round-robin flat) is preserved. Node separation shifts the curve down by a roughly constant fraction rather than changing which trend dominates. A live two-node run (server on one node, dashboard and streaming client on another, real cross-node gRPC traffic) additionally confirmed the dashboard updates correctly in real time under genuine network separation, and the final analytics matched the oracle exactly.

4. Plots

All plots are regenerated directly from the CSVs referenced above — no measured value is altered in any plot; only labels, titles, and presentation were changed for clarity.

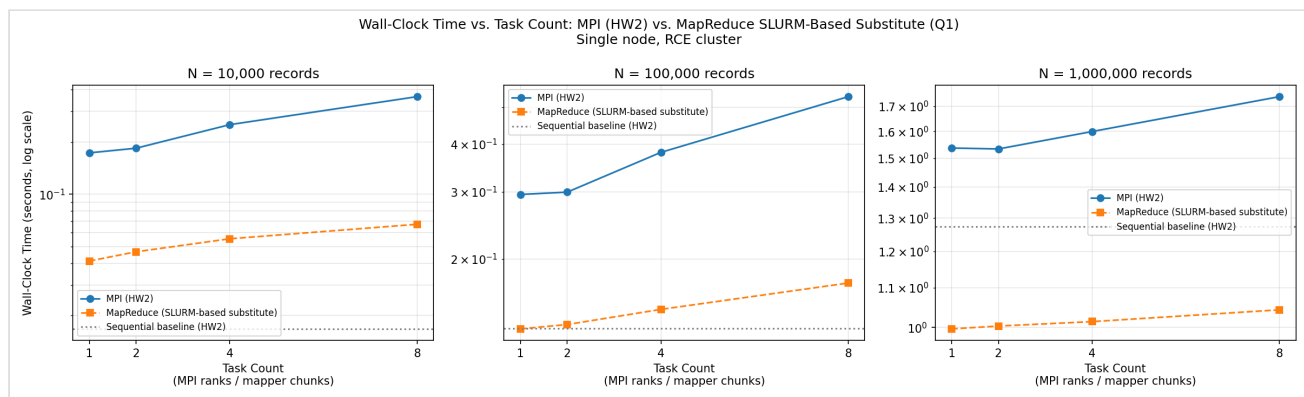


Figure 1. Wall-clock time vs. task count, one panel per input size. Log-scale y-axis; the dotted line marks the HW2 sequential baseline for reference.

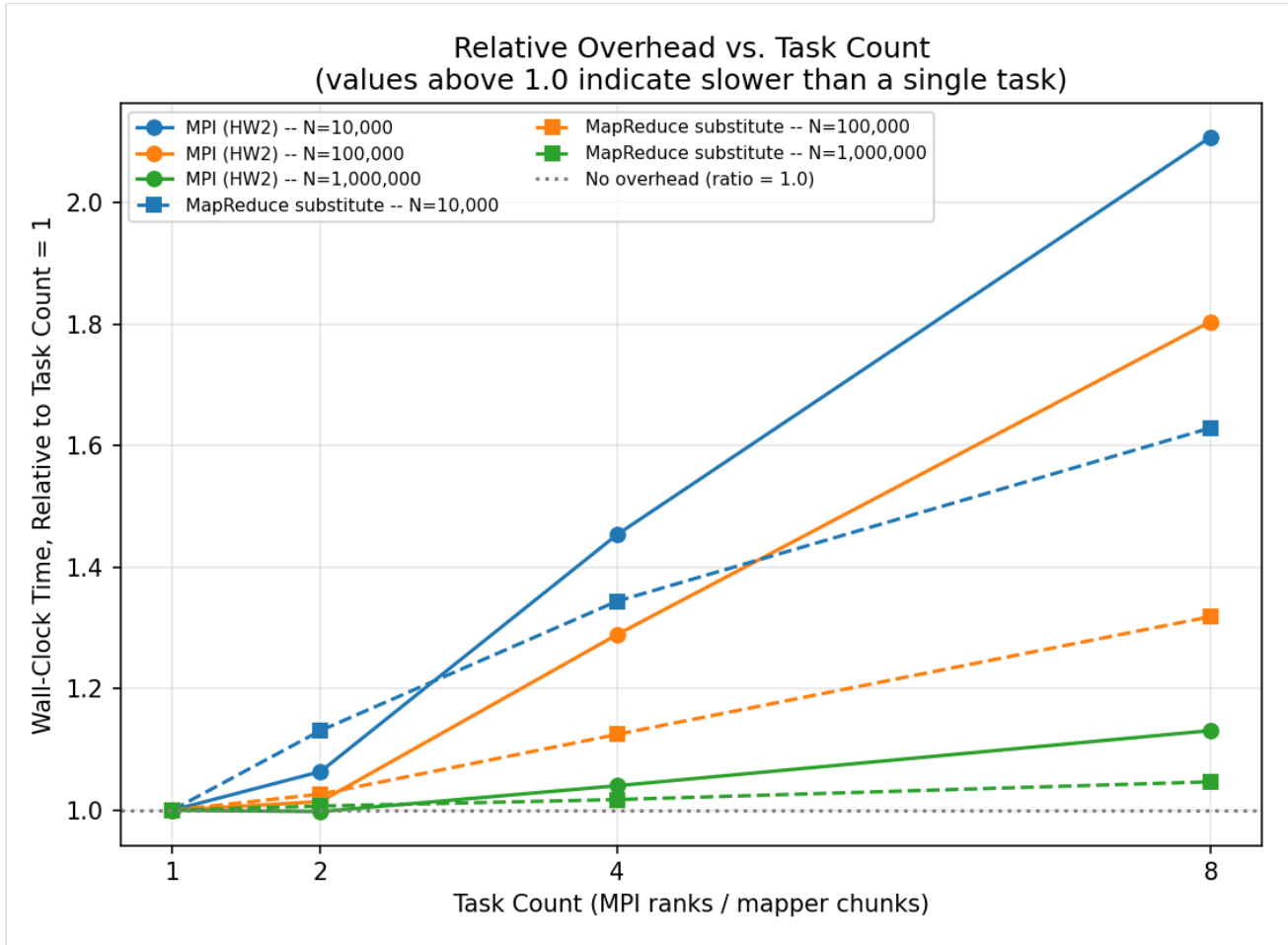


Figure 2. Each system's own wall-clock time relative to its task-count = 1 run, across all three input sizes. Values above 1.0 indicate a task count that is slower than a single task.

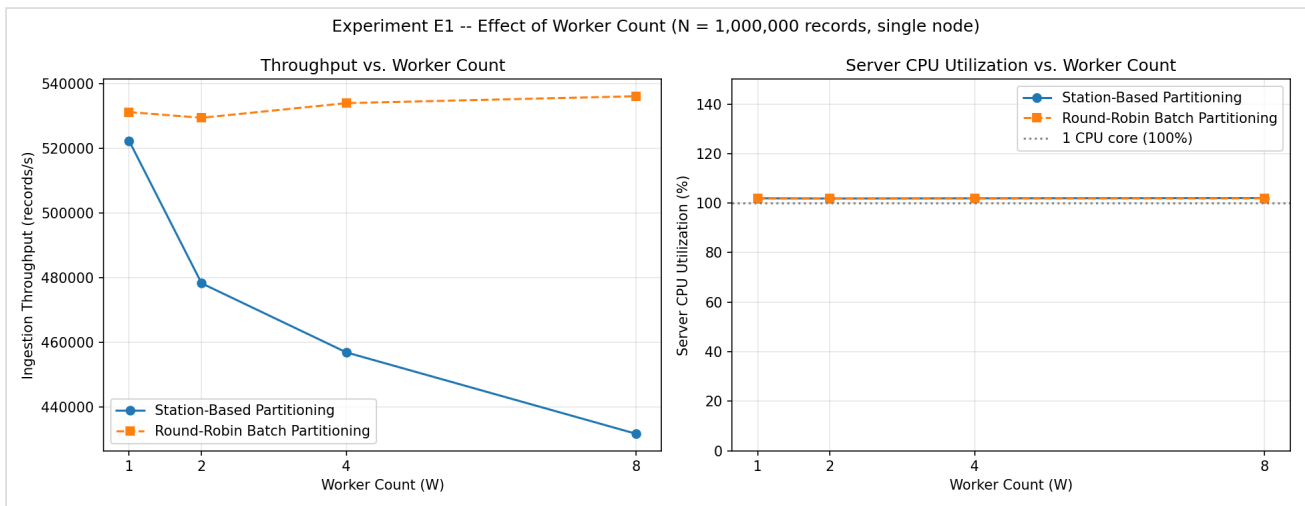


Figure 3. Ingestion throughput and server CPU utilization vs. worker count, both partition strategies. The dotted line at 100% marks one CPU core.

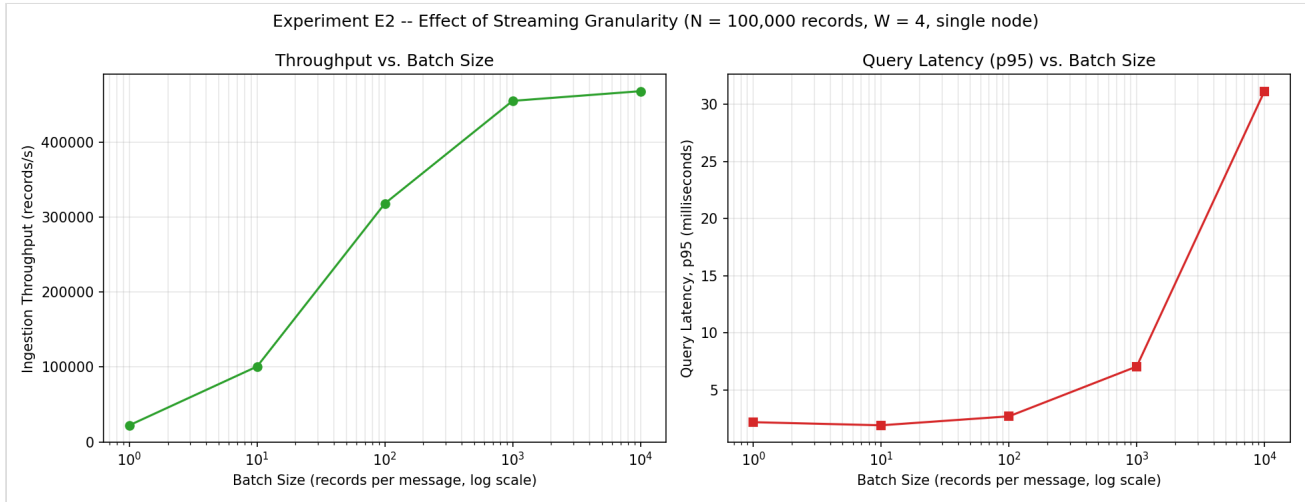


Figure 4. Throughput and query p95 latency vs. batch size (streaming granularity). X-axis is log-scale.

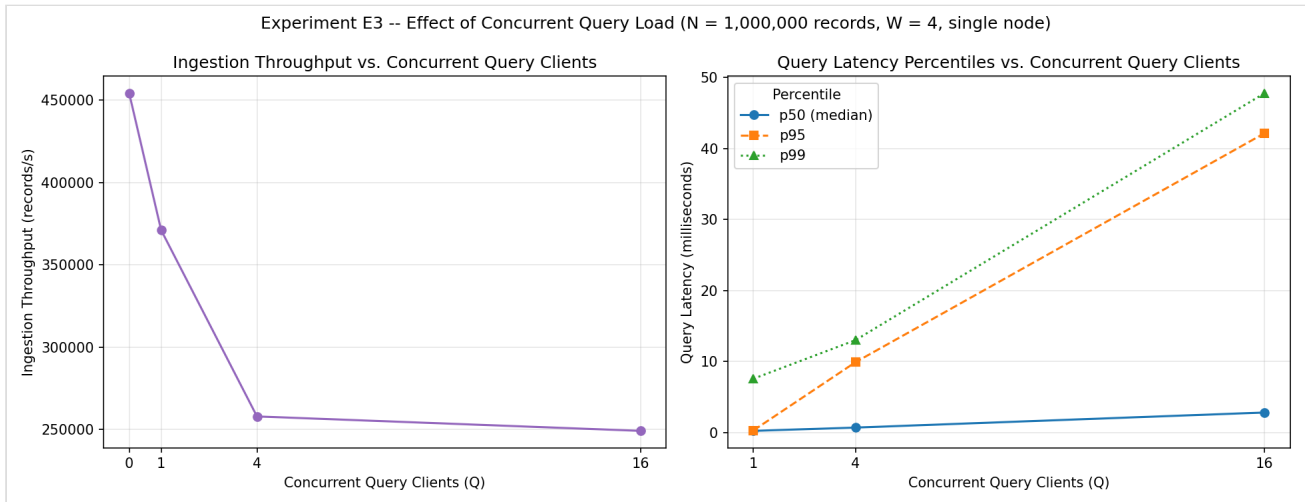


Figure 5. Ingestion throughput and query latency percentiles (p50/p95/p99) vs. concurrent query client count.

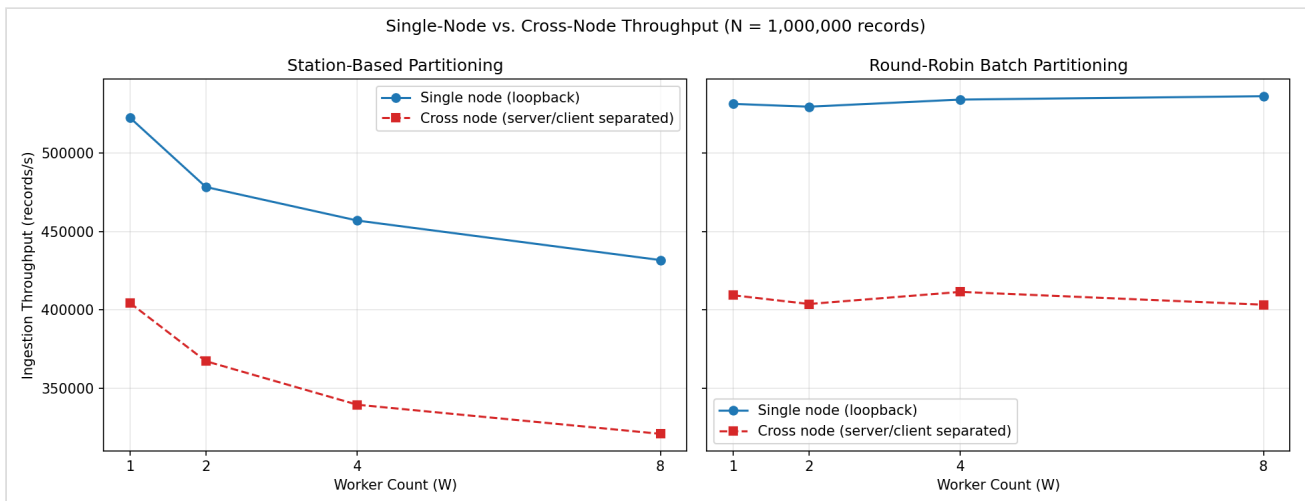


Figure 6. Single-node vs. cross-node throughput, both partition strategies, same worker-count sweep as Figure 3.

5. Q1 vs. HW2 MPI Comparison [Required for Q1]

Data sources: HW2's own real MPI benchmark (`analysis_detail.csv`) and this project's real SLURM-based substitute benchmark (job 98781), both single-node RCE runs. Matching record counts, not byte-identical datasets (HW2 used different K/S/seed per size than this project) — an "as delivered" comparison at equal input sizes, the same standard HW2's own report already used for itself.

Records	HW2 sequential	HW2 MPI (P=1/2/4/8)	Q1 SLURM-based substitute (M=1/2/4/8)
10,000	0.0168 s	0.173 / 0.184 / 0.252 / 0.365 s	0.041 / 0.047 / 0.055 / 0.067 s
100,000	0.1317 s	0.295 / 0.300 / 0.381 / 0.533 s	0.131 / 0.135 / 0.148 / 0.173 s
1,000,000	1.2737 s	1.537 / 1.534 / 1.599 / 1.739 s	0.996 / 1.003 / 1.014 / 1.043 s

* Correctness FAIL, confirmed benign precision tie (Section 2.4); times are real.

[Interpretation] MPI never beats its own sequential baseline at any tested size (best speedup 0.83× at P=2/1,000,000 records) — HW2's own finding, confirmed again here, attributed to process-launch/MPI-init overhead dominating a cheap per-record computation. The SLURM-based substitute is faster in absolute terms at every overlapping size, but this is **not** a fair "MapReduce beats MPI" conclusion: the substitute skips real disk-mediated cluster shuffle (a real Hadoop job would incur genuine cross-node data movement this single-node pipeline never does) and skips MPI's process-launch/collective-communication cost entirely. The comparison that would isolate the paradigms' inherent cost — real Hadoop/YARN vs. MPI — remains untestable until the Hadoop/YARN environment issue is resolved. Both systems show the same qualitative shape: a roughly fixed per-task cost dominates at low N and matters proportionally less as N grows (HW2 MPI's 8-task/1-task ratio shrinks from 2.1× at N=10,000 to 1.13× at N=1,000,000; the substitute's shrinks from 1.6× to 1.05× over the same range).

Qualitative comparison:

Factor	HW2 MPI	Q1 SLURM-based substitute
Work division	Rank reads a line-aligned byte range of the shared input	Driving script splits records into M chunks by line count
Intermediate state	Local per-rank aggregates, combined via MPI collectives	Per-mapper key-value rows, materialized to disk, sorted, reduced
Communication/data movement	In-memory scalar reductions and gathered summaries	Disk-backed intermediate files plus a sort pass; a real Hadoop job would additionally shuffle over the network across nodes
Task launch overhead	One <code>mpirun</code> launch for all ranks	One process launch per mapper chunk plus the reducer, no framework-managed scheduling
Programming model	Manual partitioning and collectives tightly coupled to MPI's API	Key-value emit/combine maps directly onto the real Hadoop Streaming contract — the mapper/reducer code is unmodified from what a real <code>hadoop jar</code> invocation would run

6. Observations and Limitations

- **Neither Q1's substitute nor Q2's worker threads demonstrate genuine multi-core parallelism from their respective "task count" knobs**, each for a specific, measured reason: Q1's mapper chunks execute sequentially on one core by construction of the permitted substitute; Q2's worker threads are bounded by Python's GIL (confirmed via measured CPU utilization staying at ~one core regardless of worker count). HW2's own MPI, by contrast, uses real OS processes, but its speedup is still capped by process-launch and communication overhead at these problem sizes.
- **The N=100,000 precision tie is the single most important correctness finding in this report**, because it independently reproduces in two unrelated implementations (a two-level Kahan sum in C++, a Python double-precision Kahan sum) from the same underlying exact halfway value in the dataset — strong evidence it is a genuine floating-point property, not a bug in either pipeline.
- **Q2's cross-node throughput cost (~23–26%) is a real, measured, first-class result**, not a footnote — it directly answers whether physical node separation matters for this system, rather than assuming either that it doesn't (same-node loopback) or that it dominates (untested multi-node).
- **Limitations, stated directly**: real Hadoop/YARN execution was not possible given the documented cluster environment issue, so the Q1/MPI comparison cannot yet isolate each paradigm's inherent cost from its execution substitute. Memory (RSS) usage was not instrumented for either question (listed as "where appropriate" in the assignment, not required). Streaming rate and station-count/skew were not run as standalone experiments (the mechanisms exist; running them as full

experiments was out of scope given time constraints). RCE benchmarks are 3 repetitions on a shared, possibly-contended cluster node — real and internally consistent, but not isolated-hardware confidence intervals.

7. Conclusion

Both Q1 and Q2 implement the complete HW2 Q8 analytics under their respective paradigms, verified correct against the same sequential oracle, with the one confirmed discrepancy (a genuine floating-point halfway tie at $N=100,000$) independently explained in both. Q1's real benchmark results, compared directly against HW2's own real MPI numbers, show both systems dominated by fixed per-task overhead at these problem sizes, with the caveat that Q1's numbers reflect a course-sanctioned SLURM-based substitute rather than real Hadoop/YARN execution. Q2's real benchmark results answer all three required/listed performance questions — worker count, streaming granularity, and concurrent query load — with the GIL identified as the specific, measured mechanism limiting worker-count scaling, and node separation shown to have a real, quantified throughput cost. The CLI dashboard satisfies Q2's requirement directly, verified both in automated correctness checks and in live interactive runs, including with the server and client on physically separate cluster nodes.